ASAP CRN Cloud PMDBS scRNAseq Collection

README - currently release v2.0.0, 10.5281/zenodo.14373048

Overview

The ASAP CRN Cloud is pleased to expand the **PMDBS Single Cell (sc) RNAseq Collection** as part of the v2.0.0 ASAP CRN Cloud release.

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See CRN Cloud Authorship List for the full list of contributing investigators/researchers by dataset.

ASAP CRN GitHub: github.com/ASAP-CRN
Data Release Date: 9/11/24
ASAP CRN Cloud Release Number: v2.0.0
Release DOI: 10.5281/zenodo.14270014
Curated Data File Manifest: ASAP CRN Cloud File Manifest - v3.0

PMDBS scRNAseq Collection

The raw scRNAseq data has been processed and harmonized to create a harmonized database of gene expression data. The platformed data is summarized below as Raw Data, Curated Data, and QC Summaries below. Code, explanations, and schematics for processing the platformed data can be found at the ASAP-CRN pmdbs-sc-rnaseq-wf repo.

Collection details

PMDBS scRNAseq Workflow Release - v2.2.0

- Release date: December 3, 2024
- GitHub release link
- Pipeline version: v2.2.0

Collection Datasets list

- `team-hafler-pmdbs-sn-rnaseq-pfc`
- `team-lee-pmdbs-sn-rnaseq`
- `team-hardy-pmdbs-sn-rnaseq`
- `team-scherzer-pmdbs-sn-rnaseq-mtg`
- `team-jakobsson-pmdbs-sn-rnaseq`
- `cohort-pmdbs-sc-rnaseq.` This *cohort* dataset includes samples across five contributed snRNAseq datasets above

Curated Data

Curated data is the result of processing the raw dataset contributions by ASAP CRN Workflows.

For the scRNAseq Datasets the curated data can be categorized as "*preprocessed*" and "*processed*", and "*integrated*".

Preprocessed. Pre-processing refers to the first stage of sequence alignment from the *raw* fastq files with cellranger for each sample, followed by basic QC. This QC consists count correction due to ambient RNA molecules and random barcode swapping with `cellbender`, and removing doublets with `scrublet`, yielding anndata objects for each sample (*.adata_object.h5ad) containing the full QC-ed gene expression matrix for each sample.

Processed. Processing refers to *filtering* the low quality cells – the next level of QC – and *feature selection*, yielding two key artifacts: an unprocessed (but QC-ed) data object containing all cells from all samples (e.g. *merged_adata_object.h5ad) and a final processed anndata object containing the gene expression of highly variable genes for all cells (e.g. *final_adata_object.h5ad) and as per the ASAP-CRN pmdbs-sc-rnaseq-wf. The filtering cutoffs were chosen to be:

- Mitochondrial gene percentage < 10%
- Doublet_scores < 0.2
- Total counts between 500, and 100,000

- Number features per cell between 300 and 10,000

Feature selection found the top genes with the highly_variable_genes method from Pearson's residuals on the pre-processed counts.

Integrated. The integrated or *harmonized* data represents batch-corrected aggregated data. Batch correction and integration of the individual samples into a single count table is the critical step of harmonization. In the workflow we call this entire process the "cohort analysis", and the key steps of integration take place in the "clustering" workflow.

The full *integration* is achieved through the following steps:

1. Batch correction on the aggregate data with scVI method, and
2. Clustered and annotated by creating embeddings neighbor graph), clustering on the neighbor graph at multiple Leiden resolutions, finding a UMAP embedding for visualization, and annotation with CellAssign. Cell type assignment is done with a very simple taxonomy and marker genes
3. An additional Harmony integration step is performed to make alternative batch equalization to `scVI`, which also enables quantitative assessment of the batch related variance with respect to preservation of biological variance (via `scib-metrics`)

QC Summaries. (Plots)

As advertisements to the harmonized data a family of plots illustrating the Quality Control (QC) statistics and an overall illustration of the dataset. Embedding plots showcasing the distribution of batch labels, doublet scores, cell types, total Count, N Feature, percent-mitochondria, percent-ribosome, and samples are generated. As are a scatter of UMI count vs Gene count, and violin plots for the preprocessing QC metrics.

Final Artifacts

The harmonized data is benchmarked with scib-metrics tools to characterize the quality of biological conservation and batch equalization. A visual summary and table are available (e.g. *..scib_report.csv) The "output" artifacts have "counts", "normalized counts", PCs, scVI embeddings, and Harmony equalized PCs available (e.g. `*.harmony_integrated.h5ad`.)

Raw Data

The *raw* data refers to the fastq files transferred by each ASAP CRN Team. These data are available with the caveat that they are stored in "requester pays" buckets on the google cloud platform. A google cloud billing account will need to be registered to access these data. More information is available HERE. gs://`asap-raw-<team_name>-<source>-<dataset_name>`

Metadata & Data Dictionary

Data Dictionary: ASAP CRN Cloud Data-Dictionary - v3.0

The ASAP CRN Common Data Elements (CDE - v3.0) outlines the variables which are harmonized across the ASAP CRN. These metadata enable the harmonization of the data submitted by multiple teams. The metadata consists of five tables, which are described below, and also detailed as Data Dictionaries for each.

Metadata refers to descriptive information about the overall study, individual samples, all protocols, and references to processed and raw data file names. Metadata was aggregated using the table-based submission for each team's dataset. The tables - STUDY, PROTOCOL, SUBJECT, SAMPLE, DATA ,etc. - can be thought of as spreadsheets, and each table is prepared as simple .csv files which constitute the tables outlined in the ASAP CRN CDE (final sheet). These have been aggregated across submissions and unique ASAP IDs generated for each dataset, team, subject and sample. I.e. ASAP_dataset_id, ASAP_team_id, ASAP_subject_id, and ASAP_sample_id.

The metadata can be browsed on ASAP CRN Cloud *Explorer* and exposed in Verily Workbench.

This document.